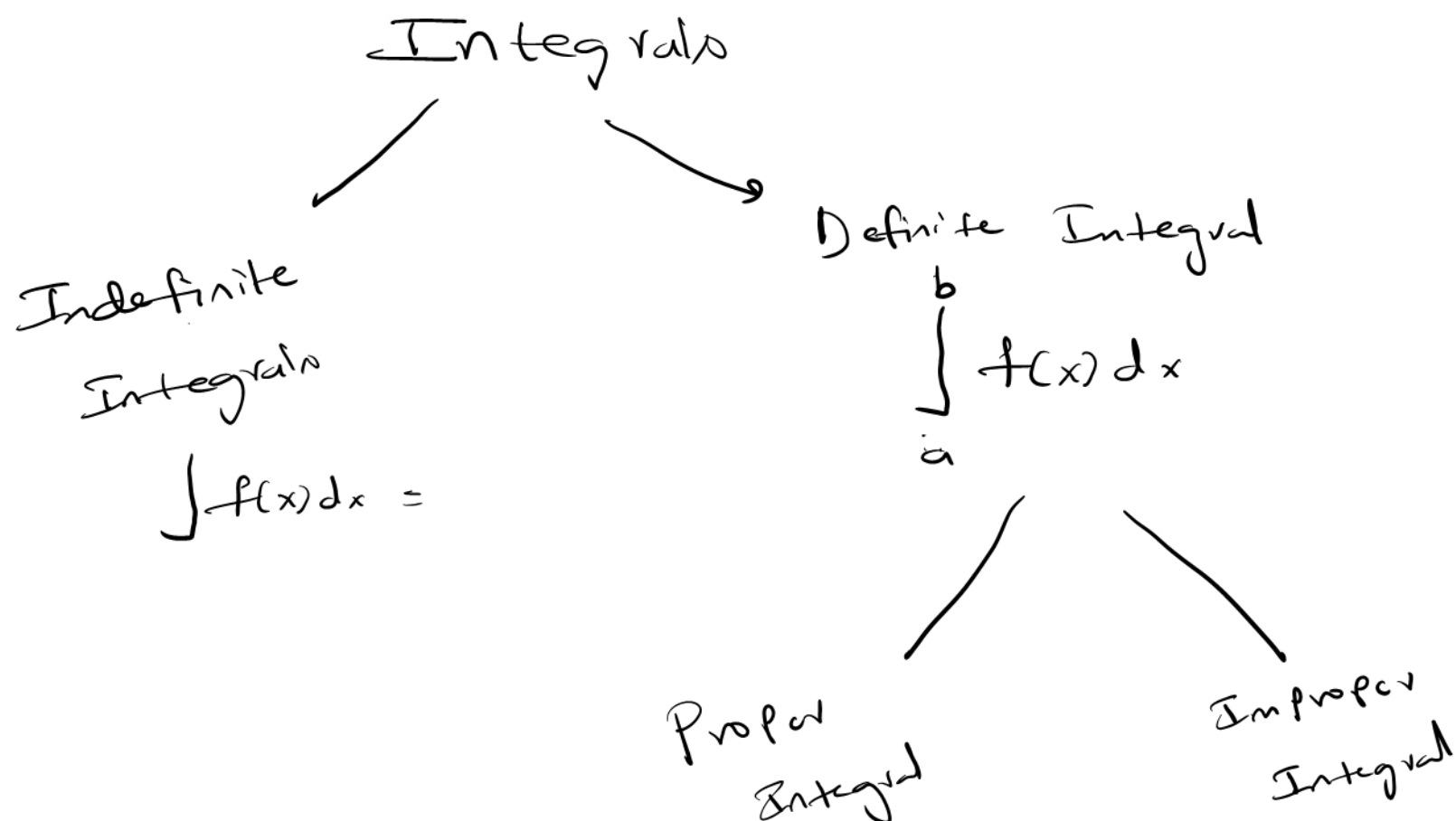


Unit-3

(Calculus)

Improper IntegralsProper Integral

If the range (limits) are finite and integrand is bounded

ex:-

$$\int_2^4 \frac{1}{2^x} dx$$

↙ ↘

finite Integrand - bounded

→ Improper Integral① First kind of Improper Integral

One (or) both the limits are not finite.

i.e. $a = -\infty$ (or) $b = +\infty$ (or) both.

ex:-

$$\int_1^{\infty} \frac{1}{x} dx, \quad \int_{-\infty}^{-1} \frac{1}{x^2} dx, \quad \int_0^{\infty} \sin x dx$$

II Improper Integral of Second kind

Both the limits of integration are finite but function $f(x)$ is not continuous (discontinuous or unbounded) at one or more points in $[a, b]$

$$\int_a^b f(x) dx$$

ex:- $\int_0^2 \frac{1}{x-1} dx$

at $x=1$

$$\frac{1}{x-1} = \frac{1}{0} = \text{not defined}$$

III Improper Integral of 3rd kind

If above two conditions are satisfied then they are called Improper integrals of 3rd kind

(i) one (or) both limits are not finite

(ii) $f(x)$ is not continuous in $[a, b]$

ex:- $\int_0^\infty \frac{1}{(1-x)^2} dx$

→ ① $\int_{-\infty}^\infty e^{-x^2} dx \rightarrow$ Improper Integral

④ $\int_{-\infty}^5 \frac{1}{x} dx \rightarrow$ Improper Integral of 3rd kind

② $\int_0^2 \sqrt{x^2+1} dx \rightarrow$ Proper

③ $\int_2^3 \frac{1}{(x-2)^3} dx \rightarrow$ Improper 2nd kind

Imp proper Integrals of INL kind

$$(a) \int_a^\infty f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx$$

$$(b) \int_{-\infty}^b f(x) dx = \lim_{R \rightarrow -\infty} \int_R^b f(x) dx$$

$$(c) \int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx$$

$$= \lim_{R_1 \rightarrow -\infty} \int_{R_1}^a f(x) dx + \lim_{R_2 \rightarrow \infty} \int_a^{R_2} f(x) dx \longrightarrow (1)$$

$$\int_{-\infty}^\infty f(x) dx = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx \longrightarrow (2)$$

→ The value of $\int_{-\infty}^\infty f(x) dx$ using equation (1) is known as general value

→ The value of $\int_{-\infty}^\infty f(x) dx$ using equation (2) is known as Principal value

→ Principal value may exist even when general value does not exist.

→ If the general value exists then Principal value = general value

If limit in R.H.S
exists we say
Improper Integral
converges otherwise
it diverges

ex:-

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x} dx$$

$$= \lim_{R \rightarrow \infty} \ln x \Big|_1^R = \lim_{R \rightarrow \infty} \ln R - \ln 1$$

$$= \lim_{R \rightarrow \infty} \ln R$$

diverges

② $\int_1^{\infty} \frac{1}{x^2} dx$

$$\int x^{-2} = \frac{x^{-2+1}}{-2+1}$$

$$= \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x^2} dx = \lim_{R \rightarrow \infty} \left[-\frac{1}{x} \right]_1^R$$

$$= \lim_{R \rightarrow \infty} \left[-\frac{1}{R} + 1 \right]$$

$$= 0 + 1 = 1$$

∴ Converges

ex:-

$$\int_{-\infty}^{\infty} x dx = \int_{-\infty}^a x dx + \int_a^{\infty} x dx$$

$$= \lim_{R_1 \rightarrow \infty} \int_{R_1}^a x dx + \lim_{R_2 \rightarrow \infty} \int_a^{R_2} x dx$$

$$= \lim_{R_1 \rightarrow \infty} \left[\frac{x^2}{2} \right]_{R_1}^a + \lim_{R_2 \rightarrow \infty} \left[\frac{x^2}{2} \right]_a^{R_2}$$

$$= \lim_{R_1 \rightarrow \infty} \left[\frac{a^2}{2} - \frac{R_1^2}{2} \right] + \lim_{R_2 \rightarrow \infty} \left[\frac{R_2^2}{2} - \frac{a^2}{2} \right]$$

$\swarrow$ diverge $\rightarrow \infty$ $\swarrow$ diverge $\rightarrow \infty$

general value is
not finite
general value
doesn't exist
∴ diverge

$$\int_{-\infty}^{\infty} x \, dx = \lim_{R \rightarrow \infty} \int_{-R}^R x \, dx$$

$$= \lim_{R \rightarrow \infty} \left[\frac{x^2}{2} \right]_{-R}^R = \lim_{R \rightarrow \infty} \left(\frac{R^2}{2} - \frac{R^2}{2} \right)$$

$$= \lim_{R \rightarrow \infty} 0 = 0$$

0 is the principal value

$$\int_a^b f(x) dx \rightarrow$$

1st kind improper integral

→ 2nd kind →

→ 3rd kind

1st kind

$$\int_{-\infty}^b f(x) dx = \lim_{R \rightarrow -\infty} \int_R^b f(x) dx$$

$$\int_a^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx$$

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx$$

$$= \lim_{R_1 \rightarrow -\infty} \int_{R_1}^a f(x) dx + \lim_{R_2 \rightarrow \infty} \int_a^{R_2} f(x) dx \rightarrow \textcircled{1}$$

general value

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx \rightarrow \text{Principal value}$$

Improper Integrals of 2nd Kind

$\int_a^b f(x) dx$: $f(x)$ is not defined at one or more points in $[a, b]$

→ If $f(x)$ is not defined at point b

$$\textcircled{1} \int_a^b f(x) dx = \lim_{t \rightarrow b} \int_a^t f(x) dx$$

$$= \lim_{\varepsilon \rightarrow 0} \int_a^{b-\varepsilon} f(x) dx$$



→ If $f(x)$ is not defined at point a

$$\int_a^b f(x) dx = \lim_{t \rightarrow a} \int_t^b f(x) dx$$

$$= \lim_{\varepsilon \rightarrow 0} \int_{a+\varepsilon}^b f(x) dx$$



→ If $f(x)$ is not defined at point c $a < c < b$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$= \lim_{\varepsilon \rightarrow 0} \int_a^{c-\varepsilon} f(x) dx + \lim_{\varepsilon' \rightarrow 0} \int_{c+\varepsilon'}^b f(x) dx \rightarrow \textcircled{1}$$

↓
Cauchy value

→ $\varepsilon = \varepsilon'$

$$\int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0} \left\{ \int_a^{c-\varepsilon} f(x) dx + \int_{c+\varepsilon}^b f(x) dx \right\} \rightarrow \text{Principal value}$$

→ We can compare situation when ε and $\varepsilon' \rightarrow 0$
 Independently limits may not exist. but if $\varepsilon = \varepsilon'$ then
 if limits exist. Then we have Principal value

ex 1:- $\int_0^1 \frac{1}{\sqrt{1-x}} dx$ → Improper Integral of 2nd kind

when $x=1$ $f(x) = \frac{1}{\sqrt{1-x}}$ is not defined

$$\int_0^1 \frac{1}{\sqrt{1-x}} dx = \lim_{\varepsilon \rightarrow 0} \int_0^{1-\varepsilon} \frac{1}{\sqrt{1-x}} dx = \int_0^{1-\varepsilon} (1-x)^{-1/2} dx$$

$$= \left[\frac{(1-x)^{-1/2+1}}{-\frac{1}{2}+1} \right] = \lim_{\varepsilon \rightarrow 0} -2 \left[1-x \right]^{1/2} \Big|_0^{1-\varepsilon}$$

$$= \lim_{\varepsilon \rightarrow 0} -2 \left[(1-1+\varepsilon)^{1/2} - 1 \right] = \lim_{\varepsilon \rightarrow 0} -2 \left[\varepsilon^{1/2} - 1 \right]$$

= 2 //

ex 2 → $\int_{-1}^1 \frac{1}{x} dx$ → Improper Integral of kind 2.

when $x=0$ $f(x) = \frac{1}{x}$ is not defined

$$\int_{-1}^1 \frac{1}{x} dx = \int_{-1}^0 \frac{1}{x} dx + \int_0^1 \frac{1}{x} dx$$

$$= \lim_{\varepsilon \rightarrow 0} \int_{-1}^{0-\varepsilon} \frac{1}{x} dx + \lim_{\varepsilon' \rightarrow 0} \int_{0+\varepsilon'}^1 \frac{1}{x} dx$$

$$= \lim_{\varepsilon \rightarrow 0} \ln|x| \Big|_{-1}^{-\varepsilon} + \lim_{\varepsilon' \rightarrow 0} \ln|x| \Big|_{\varepsilon'}^1$$

Since diverging
 general value
 doesn't exist

$$= \lim_{\varepsilon \rightarrow 0} \ln|-\varepsilon| - \ln|-1| + \lim_{\varepsilon' \rightarrow 0} \ln|1| - \ln|\varepsilon'| \rightarrow \infty$$

diverging

Principal value

$\varepsilon = \varepsilon'$

$$\int_{-1}^1 f(x) dx = \lim_{\varepsilon \rightarrow 0} \left[\int_{-1}^{-\varepsilon} \frac{1}{x} dx + \int_{\varepsilon}^1 \frac{1}{x} dx \right]$$

$$= \lim_{\varepsilon \rightarrow 0} \left[\ln|x| \Big|_{-1}^{-\varepsilon} + \ln|x| \Big|_{\varepsilon}^1 \right]$$

$$= \lim_{\varepsilon \rightarrow 0} \left[\ln|-\varepsilon| - \ln|-1| + \ln|1| - \ln|\varepsilon| \right]$$

$$= \lim_{\varepsilon \rightarrow 0} \left[\ln \cancel{\varepsilon} - \ln|1| + \ln|1| - \ln \cancel{\varepsilon} \right] = 0 \quad \text{Principal value}$$

0 0

ex:-

$$\int_{-1}^3 \frac{dx}{(x-2)^3}$$

Improper ^{Interval} of 2nd kind

$$f(x) = \frac{1}{(x-2)^3} \text{ is not defined when } x=2$$

$$\rightarrow \int_{-1}^3 \frac{1}{(x-2)^3} dx = \int_{-1}^2 \frac{1}{(x-2)^3} dx + \int_2^3 \frac{1}{(x-2)^3} dx$$

$$= \lim_{\varepsilon \rightarrow 0} \int_{-1}^{2-\varepsilon} \frac{1}{(x-2)^3} dx + \lim_{\varepsilon' \rightarrow 0} \int_{2+\varepsilon'}^3 \frac{1}{(x-2)^3} dx$$

$$= \lim_{\varepsilon \rightarrow 0} \left[\frac{-1}{2(x-2)^2} \right]_{-1}^{2-\varepsilon} + \lim_{\varepsilon' \rightarrow 0} \left[\frac{-1}{2(x-2)^2} \right]_{2+\varepsilon'}^3$$

$$= \lim_{\varepsilon \rightarrow 0} \left[\frac{-1}{2\varepsilon^2} + \frac{1}{2(-3)^2} \right] + \lim_{\varepsilon' \rightarrow 0} \left[\frac{-1}{2(\varepsilon')^2} + \frac{1}{2(\varepsilon')^2} \right]$$

diverges

diverges

diverges

So Grand value doesn't exist

Principal value

$$\boxed{\Sigma = \Sigma'}$$

$$\lim_{\Sigma \rightarrow 0} \left[-\frac{1}{\cancel{2}\Sigma^2} + \frac{1}{18} - \frac{1}{\Sigma} + \frac{1}{\cancel{2}\Sigma^2} \right]$$

$$\lim_{\Sigma \rightarrow 0} \left[\frac{1}{18} - \frac{1}{\Sigma} \right] = -\frac{4}{9} \rightarrow \text{Principal value}$$

* NoteIf $f(x)$ is discontinuous at both ends of theinterval $[a, b]$

$$\int_a^b f(x) dx = \int_{(a)}^c + \int_c^{(b)}$$

$$= \lim_{\epsilon \rightarrow 0} \int_{a+\epsilon}^c + \lim_{\epsilon' \rightarrow 0} \int_c^{b-\epsilon'}$$

→ Convergence of Improper Integrals

→ Standard Results

$$\int_1^{\infty} \frac{1}{x^p} dx$$

$$= \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x^p} dx = \lim_{R \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^R$$

$$= \lim_{R \rightarrow \infty} \left[\frac{R^{-p+1}}{1-p} - \frac{1^{-p+1}}{1-p} \right]$$

$$\frac{1}{R}$$

$$= \lim_{R \rightarrow \infty} \left[\frac{R^{1-p}}{1-p} - \frac{1}{1-p} \right]$$

→ Converges
if $p > 1$

$$\rightarrow \int_1^{\infty} \frac{1}{x^p} dx$$

It converges when $p > 1$

diverges when $p \leq 1$

$$p=1$$

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{R \rightarrow \infty} \int_1^R \frac{1}{x} dx = \lim_{R \rightarrow \infty} \ln x \Big|_1^R$$

$$\lim_{R \rightarrow \infty} = \ln R - \ln 1$$

diverging

Some standard results

①

$$\int_1^{\infty} \frac{1}{x^p} dx$$

converges $p > 1$

diverges $p \leq 1$

$$\int_a^{\infty} \frac{1}{x^p} dx$$

②

$$\int_0^a \frac{1}{x^p} dx$$

converges $p < 1$

diverges $p \geq 1$

③

$$\int_a^b \frac{1}{(x-a)^p} dx$$

converges for $p < 1$

diverges for $p \geq 1$

④

$$\int_a^b \frac{dx}{(b-x)^p}$$

converges for $p < 1$

diverges for $p \geq 1$